MIL-R-23767B(AS)
13 November 1970
SUPERSEDING
MIL-R-23767A(AS)
15 October 1967

MILITARY SPECIFICATION

RECEIVING SET, RADIO AN/ARR-69 AND AN/ARR-69A

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide one-way radio communication with properly equipped surface stations and other aircraft by means of amplitude modulated radio telephony. When installed with the necessary associated equipment, the Radio Set shall be suitable for the following uses:

(1) Reception of voice communications

(2) Reception of special type signals consistent with performance requirements outlined herein.

(3) Automatic direction finding when installed with certain equipment.

* 1.2 Classification - The equipment covered by this specification shall consist of the following items:

TYPE DESIGNATION	APPLICABLE PARAGRAPH
R-1286/ARR-69	3.5.1
R-1286A/ARR-69A	3.5.1
MT-3137/ARR-69	3.5.2
MT-3590/ARR-69	3.5.2

* 1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

* 2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-M-81288	Mounting Bases; Flexible Plastic Foam
MIL-F-81334	Foam, Plastic, Flexible, Open Cell Polyester Type, Polyurethane

STANDARD

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

* 2.2 Availability of Documents -

When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

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3. REQUIREMENTS

3.1 Production - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies, without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) shall be obtained as outlined in Specification MIL-E-5400.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 9.5 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 1000 hours of mean (operating) time between failures when

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tested and accepted as outlined under the requirements of 4. 4. 2.

* 3. 3. 2. 4 Time Totalizing Meter - The following units shall contain a time totalizing meter in accordance with Specification MIL-M-7793.

Unit	Type of Meter
R-1286/ARR-69 and R-1286A/ARR-69A	MS17321-2

3. 3. 3 Cabling and Connections -

3. 3. 3. 1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3. 3. 3. 2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provide the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3. 3. 4 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3. 3. 5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3. 3. 6 Maintenance Provisions and Field Testing - Provisions for maintenance and field testing shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

* 3. 3. 7 Nomenclature and Nameplates - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3. 3. 8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

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Temperature	Room ambient (25±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5±0.5VDC

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combinations of these conditions as specified in Specification MIL-E-5400 for Class 3 equipment, except as modified herein:

3.3.9.1 Vibration - The equipment with vibration isolators in place shall operate satisfactorily while being subjected to the vibration requirements of Curve 1 of MIL-E-5400. The vibration requirement for the equipment without vibration isolators is detailed below:

Frequency	Displacement/Acceleration
5 to 33 cps	0.11 inch double amplitude
33 to 50 cps	±5 g
50 to 73 cps	±3 g
73 to 500 cps	±2 g

3.3.9.2 Temperature/Altitude - The receiver shall be capable of satisfactory operation for sixty minutes at an altitude of 80,000 feet in an ambient of 70°C without cooling air.

3.3.9.3 Humidity - The equipment shall operate satisfactorily when subjected to the Humidity Test of Specification MIL-T-5422, except that the maximum temperature shall be 71°C.

3.3.9.4 Explosive Atmosphere - The equipment shall operate satisfactorily when subjected to the explosion test, procedure 1 of Specification MIL-T-5422, except that one additional test shall be conducted at a simulated test altitude of 60,000 feet.

3.3.9.5 Sand and Dust - The equipment shall operate satisfactorily after being subjected to the sand and dust test of Specification MIL-T-5422, except that the dust velocity through the test chamber shall be 350±250 feet per minute.

3.3.9.6 Acoustic Noise - The equipment shall operate satisfactorily while exposed to a wide band noise of the overall level of 135±5 db. The source direction for the testing should be considered as being normal to the exterior structure of the aircraft. The testing period duration shall be 5 hours.

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3.3.9.7 Equipment Maneuver Load Factor (Limit) - The equipment shall operate satisfactorily when subjected to constant acceleration forces as shown herein: The test duration shall be for one minute in each of the six directions.

	Vertical-Down	Vertical-Up	Horizontal-Major & Minor Axis
Limit Load	8.5 g	3.3 g	± 6.1 g

3.3.9.7.1 Equipment Maneuver Load Factor (Ultimate) - The equipment shall withstand ultimate loads of 1.5 times the limit loads of 3.3.9.7. Bending and distortion shall be permitted, however, there shall be no failure of the attaching points and the equipment or dummy load shall remain in place.

3.3.10 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 1 minute under standard conditions and 3 minutes at extreme service conditions.

3.3.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) DC Power, 28V, Category "B", 0.5 Amp.

3.3.11.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - Unless otherwise specified values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall accept amplitude modulated UHF signals from an antenna, amplify and demodulate them, and provide audio to the aircraft interphone system or to associated automatic direction finding equipment.

3.5 Detail Requirements -

* 3.5.1 Receiver, Radio R-1286/ARR-69 and R-1286A/ARR-69A - The R-1286/ARR-69 and R-1286A/ARR-69A are identical except that channel frequencies differ. The R-1286/ARR-69 is equipped with Navy assigned

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frequencies and the R-1286A/ARR-69A is equipped with Air Force assigned frequencies. The receivers shall meet the following requirements:

3.5.1.1 Function - The receiver shall operate normally as an ADF receiver. In addition, it will provide auxiliary or emergency voice reception in case of the failure of a UHF command receiver. The operating frequency range shall be 265.0 mc to 284.9 mc with an additional guard frequency of 243.0 mc available when the 265.0 mc to 284.9 mc channels are not in use.

3.5.1.2 Form Factor - The receiver shall be contained in a single unit capable of being mounted in an aircraft by means of either mounting shown in Figure 2. It shall be possible to dismount the unit from its mounting and remove the dust cover using ordinary hand tools and without requiring removal of any cables other than the coaxial and power connectors. The receiver shall comply with the outline dimensions of Figure 1 herein.

3.5.1.3 Weight - The weight of the receiver shall not exceed 8.0 pounds.

3.5.1.4 Contents - The receiver shall contain a preselector, an RF unit, an IF and audio unit, and other parts necessary to make the unit function. It shall be possible to replace any plug-in unit with another of the same type without resynchronization except that realignment of tuning elements may be necessary.

3.5.1.5 Internal Controls - The receiver AUDIO GAIN control and the SQUELCH control shall be screwdriver adjustments that are accessible with the dust cover removed.

3.5.1.6 Electrical Connections -

3.5.1.6.1 External - The following electrical connections shall be provided on the front panel of the receiver:

Reference Designation	Receptacle Type	Function
J1	BNC (UG-657/U)	Antenna Connector
J2	Mates with MS 3116-14-19S	Power Input
J3	Mates with MS 3116-14-15P	Test Connector

* 3.5.1.7 Frequency Selection - By means of Control C-1457/ARR-40, C-4850/ASQ-85(V) or C-6684/ASQ the radio set shall be capable of reception on 20 channels with the range of 265.0 to 284.9 mc, and one guard channel within the range of 238.0 to 248.0 mc as outlined below.

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(1) By selection of a crystal of the proper frequency, Channel 1 reception shall be obtained on any one frequency within the range of 265.0 to 265.9 mc. Similarly, Channel 2 shall be selected in the range 266.0 to 266.9 mc, Channel 20 shall be selected within the range 284.0 to 284.9. Unless otherwise specified, the assignment channel frequencies shall be as follows:

(a) R-1286/ARR-69

1 - 265.2 mc	6 - 270.8 mc	11 - 275.6 mc	16 - 280.4 mc
2 - 266.6 mc	7 - 271.6 mc	12 - 276.2 mc	17 - 281.0 mc
3 - 267.6 mc	8 - 272.2 mc	13 - 277.2 mc	18 - 282.0 mc
4 - 268.6 mc	9 - 273.2 mc	14 - 278.0 mc	19 - 283.0 mc
5 - 269.8 mc	10 - 274.8 mc	15 - 279.0 mc	20 - 284.2 mc

(b) R-1286A/ARR-69A

1 - 265.5 mc	6 - 270.1 mc	11 - 275.8 mc	16 - 280.5 mc
2 - 266.6 mc	7 - 271.6 mc	12 - 276.4 mc	17 - 281.2 mc
3 - 267.6 mc	8 - 272.7 mc	13 - 277.2 mc	18 - 282.8 mc
4 - 268.6 mc	9 - 273.5 mc	14 - 278.4 mc	19 - 283.0 mc
5 - 269.9 mc	10 - 274.8 mc	15 - 279.4 mc	20 - 284.2 mc

(2) By selection of a crystal of the proper frequency, reception of one guard channel shall be obtained within the range of 238.0 to 248.0 mc. Unless otherwise specified guard channel operation shall be adjusted for 243.0 mc.

3.5.1.8 Terminations -

3.5.1.8.1 RF Terminations - The RF input to the receiver shall be through a 52 ohm cable from a signal generator having an effective source impedance of 52 ohms.

3.5.1.8.2 Audio Termination - The audio output of the receiver shall be double ended and shall be terminated in selectable 150, 300 and 600 ohm noninductive loads which shall include means for measuring audio frequency power output. The auxiliary (high fidelity) output of the receiver shall be terminated in a resistive load of 20,000 ohms, including means for measuring the output voltage.

3.5.1.9 Sensitivity - Under standard conditions the sensitivity of the receiver shall be such that an RF input of 5 open-circuit microvolts modulated 30% at 1000 cps produces an audio output of at least 50 milliwatts with a

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signal plus noise-to-noise ratio (Modulated to unmodulated) of 10 db or greater. The sensitivity (in microvolts for a 10 db signal plug noise-to-noise ratio) measured under service conditions shall not be worse than 18 microvolts. Under the reduced performance condition, a signal plus noise-to-noise ratio of at least 10 db shall be obtained when a 10,000 microvolt signal modulated 30% at 1000 cps is applied.

3.5.1.10 AVC Characteristics - The AVC action shall be such that, when a 1000 microvolt signal modulated 30% at 1000 cps is applied, the audio output power shall be 250 milliwatts (-10% + 20%); and when the signal input (30% modulated at 1000 cps) is varied from 10 microvolts to 100,000 microvolts, the audio output power shall not vary by more than ±3 db from the audio output obtained with the 1000 microvolt signal. Under service conditions the audio output shall not vary by more than ±4 db from the audio output obtained with the 1000 microvolt signal. The lower limit for AGC action is 18 microvolts. Additionally, the receiver shall be protected against blocking under conditions of input signals up to 0.5 volt.

3.5.1.11 Selectivity - The overall selectivity of the receiver shall fall within the limits set forth below:

Attenuation Below Resonant Response	Total Receiver Bandwidth
6 db	90 KC Minimum
60 db	350 KC Maximum

* 3.5.1.12 Images and Other Spurious Responses - When receiving on any channel, the receiver shall attenuate images to the maximum practicable extent and, in any case, not less than 60 db relative to the desired signal. No more than 3 spurious responses (except images) shall be attenuated between 40 to 60 db on any channel.

3.5.1.13 IF Rejection - Under standard or service conditions, the receiver shall present an attenuation of not less than 80 db, relative to the response of the desired signals to signals on the frequencies of the intermediate frequency amplifiers.

3.5.1.14 Regeneration - The interstage and overall regeneration of the receiver shall be so minimized that no evidence of unnecessary oscillation instability, or excessive regeneration shall occur at maximum RF gain setting under either standard or service conditions.

3.5.1.15 Noise Peak Limiter - The receiver shall be provided with an instantaneous noise peak limiter. The noise limiting action shall effectively limit or reduce voltages, surges, or peaks exceeding the receiver output obtained with a 45 to 65% modulated signal at maximum sensitivity.

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Its action shall be relatively instantaneous so as not to obliterate signals by a permanent reduction in receiver gain under the influence of continuous RF noises such as are caused by commutators, ignition systems, etc. When a pulsed signal having a pulse duration of 10 microseconds, a pulse repetition frequency of approximately 1000 pulses per second, and peak amplitudes up to 0.1 volt is applied on the same frequency as a desired signal modulated 30% at approximately 1000 cps, not more than 50 microvolts input (desired signal) shall be required to obtain a signal plus noise-to-noise ratio of 3 db.

3.5.1.16 Audio Characteristics - The audio characteristics of the receiver shall be as set forth below.

3.5.1.16.1 Capability - Under standard conditions, the receiver shall be capable of delivering an output of at least 250 milliwatts when a signal of 100 microvolts modulated 30% at 1000 cps is applied.

The total harmonic content of the receiver output shall not exceed 10% when a signal having any level between 100 and 100,000 microvolts modulated 30% at 1,000 cps is applied. Under any service condition, the output shall not be more than 6 db below that obtained under standard conditions and the distortion shall not exceed 20%. Under service conditions the output shall be not less than 35 milliwatts when a signal of 18 microvolts modulated at 1,000 cps is applied. Under the reduced performance conditions, an output of at least 30 milliwatts shall be obtained when a 10,000 microvolt signal modulated 30% at 1000 cps is applied.

3.5.1.16.2 Overall Fidelity - Audio response from 300 to 4,000 cps shall not vary more than +1 to -3 db relative to the response at 1000 cycles per second. Modulation frequencies outside the 300-4000 cps range shall be attenuated the maximum practicable without recourse to inclusion in the receiver of large and/or heavy filter components.

3.5.1.16.3 Auxiliary Output - The auxiliary audio output shall be in-phase ± 10° with the negative portion of the RF signal envelope when the RF input is modulated 30% at 1000 cps.

3.5.1.17 Volume Control - Operation of the volume control on the radio set control unit shall cause a minimum of change in the impedance which the radio set presents to external equipment attached to the headset terminals, and such impedance shall not be less than 50 ohms at any setting of the volume control.

3.5.1.18 RF Gain Control - The RF gain control shall be a 1000 ohm rheostat located in the control box. As the resistance of the rheostat is varied from 0 to 1000 ohms the sensitivity of the receiver shall change from 5 uv to a value between 10 and 500 uv.

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3.5.1.19 Squelch Circuit - The receiver shall be provided with a circuit which shall suppress unwanted signals or noise in the receiver output under condition of no or very low RF input signal to the receiver. It shall suppress all RF input signals below 1.5 uv modulated 30% at 1 KC and give an audio output of at least 50 mw when signals of 6 uv or greater modulated 30% at 1 KC are received. Operation of the squelch circuit shall have positive action and require at its threshold of operation, a change of approximately one decibel in RF input (30% modulated at 1 KC) for effecting a change in audio output of 10 decibels.

Under service conditions the squelch shall suppress RF signals below 1.5 uv and shall open when an RF input of 18 microvolts, 30% modulated at 1,000 cps is applied.

The time required for the squelch to close shall not exceed 2 seconds when unsquelched signals have been previously present for at least 5 seconds. When signals have been present for less than 5 seconds, the time required for the squelch to close shall not exceed 4 seconds.

3.5.1.20 Noise Modulation - Under standard conditions the signal plus noise-to-noise ratio shall be at least 30 db when a 10,000 microvolt signal modulated 30% at 1000 cps is applied and the receiver output (signal plus noise) is adjusted to 250 milliwatts.

Under service conditions of 3.3.9 the signal plus noise-to-noise ratio shall be at least 25 db when a 10,000 microvolt signal modulated 30% at 1000 cps is applied and the receiver output (signal plus noise) is adjusted to 250 milliwatts.

3.5.1.21 Channel Changing Time - The time between the instant the channel selector switch is positioned to select a given channel and the instant the equipment is in an operative condition on the selected channel shall not be greater than 3 seconds under standard conditions and not greater than 6 seconds under service conditions. This requirement shall hold for shifting the operating frequency of the equipment between any two available channels. During channel changing operations, the output of the receiver (headset circuit) shall be shorted or otherwise reduced to zero.

3.5.1.22 Accuracy of Automatic Tuning - When the same frequency (channel) is selected several times in succession, the receiver audio power (with an input signal of 100 microvolts or greater modulated 30% at 1000 cps) shall not vary by more than 0.5 db from the power output obtained initially. This requirement shall apply under standard or service conditions.

3.5.1.23 Frequency Stability - After 3 minutes of operation, the frequency stability of the receiver shall not exceed ±22.0 KC at any ambient temperature between -55°C to +125 °C. Between +125° C and +150°C the frequency stability shall not exceed ±26 KC.

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3.5.2 Mounting - A mount shall be provided. This shall be either a hard-mount (3.5.2.1) or an isolation-mount (3.5.2.2) as required by the contract.

3.5.2.1 Mounting MT-3590/ARR-69 - This mounting is of the hard-mount type with two front-located knurled nut fasteners to allow convenient removal of the receiver from the mounting.

3.5.2.1.1 Form Factor - The mounting shall comply with the dimensions outlined in Figure 2 herein. Provisions for attachment of the receiver to the mount and the mount to the aircraft shall be in accordance with Figure 2.

3.5.2.1.2 Weight - The weight of the mounting shall not exceed 0.75 pound.

3.5.2.2 Mounting MT-3137/ARR-69 - This mounting shall provide shock and vibration isolation and shall be fabricated in accordance with Specification MIL-M-81288. The flexible plastic foam material shall have a density of six pounds per cubic foot and shall be in accordance with MIL-F-81344(AS).

3.5.2.2.1 Form Factor - The mounting shall comply with the dimensions outlined in Figure 2 except that the diameter of the top plate mounting holes shall be 0.335 inch. Provisions for attachment of the receiver to the mount and the mount to the aircraft shall be in accordance with Figure 2.

3.5.2.2.2 Weight - The weight of the mounting shall not exceed 1.5 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

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(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article)Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

* 4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Initial Production Tests - One equipment shall be selected from the first 10 production equipments submitted for acceptance and sent at the contractors expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspection. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Reliability Assurance Tests
(3) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operation Test
(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 10 hour period of operation following the

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10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10 hour period specified above may be composed of two 5 hour periods to conform with standard working hours. This test shall not be required if a reliability test is selected that requires a burn in test for all equipment.

4.4.2 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.2.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.2.2 Reliability Production Acceptance Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests", except that all equipments produced shall be tested. Test Level F of MIL-STD-781 shall be used.

4.4.2.2.1 All Equipment Test - Each equipment produced shall be tested for 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment

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workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the Reliability Assurance Tests.

4.4.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the sample, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receiver at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
DC Voltage	27.0±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced

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and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concernig previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555, and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment covered by this specification shall be suitable for installation and use in all types of military aircraft and, when so installed with the necessary accessories, shall be suitable for the following uses:

(1) Reception of Voice Communications

(2) Reception of Special Type Signals

(3) Automatic Direction Finding

6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 27.5±0.5 VDC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete support information.

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6.4 Non-Repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

* 6.6 Type Designation - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Selection of applicable levels of packaging and packing (see 5.1).

(c) Type of mounting (See 1.2).

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6.8 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

Control, Receiver	C-1457/ARR-40
Control, Integrated Center	C-4850/ASQ-85(V)
Control, Radio Set	C-6684/ASQ
Direction Finder Group	AN/ARA-48 (mod) or equal
Antenna	AT-141A or equal
Headset	H-173/AR or equal
Interphone	AN/AIC-14 or equal
Interphone	AN/AIC-18 or equal
Amplifier, ADF	AM-3624/ARA-50
Intercom Station	LS-459/AIC and LS-460/AIC

6.9 Revisions - In specification revisions and superseding amendments an asterick "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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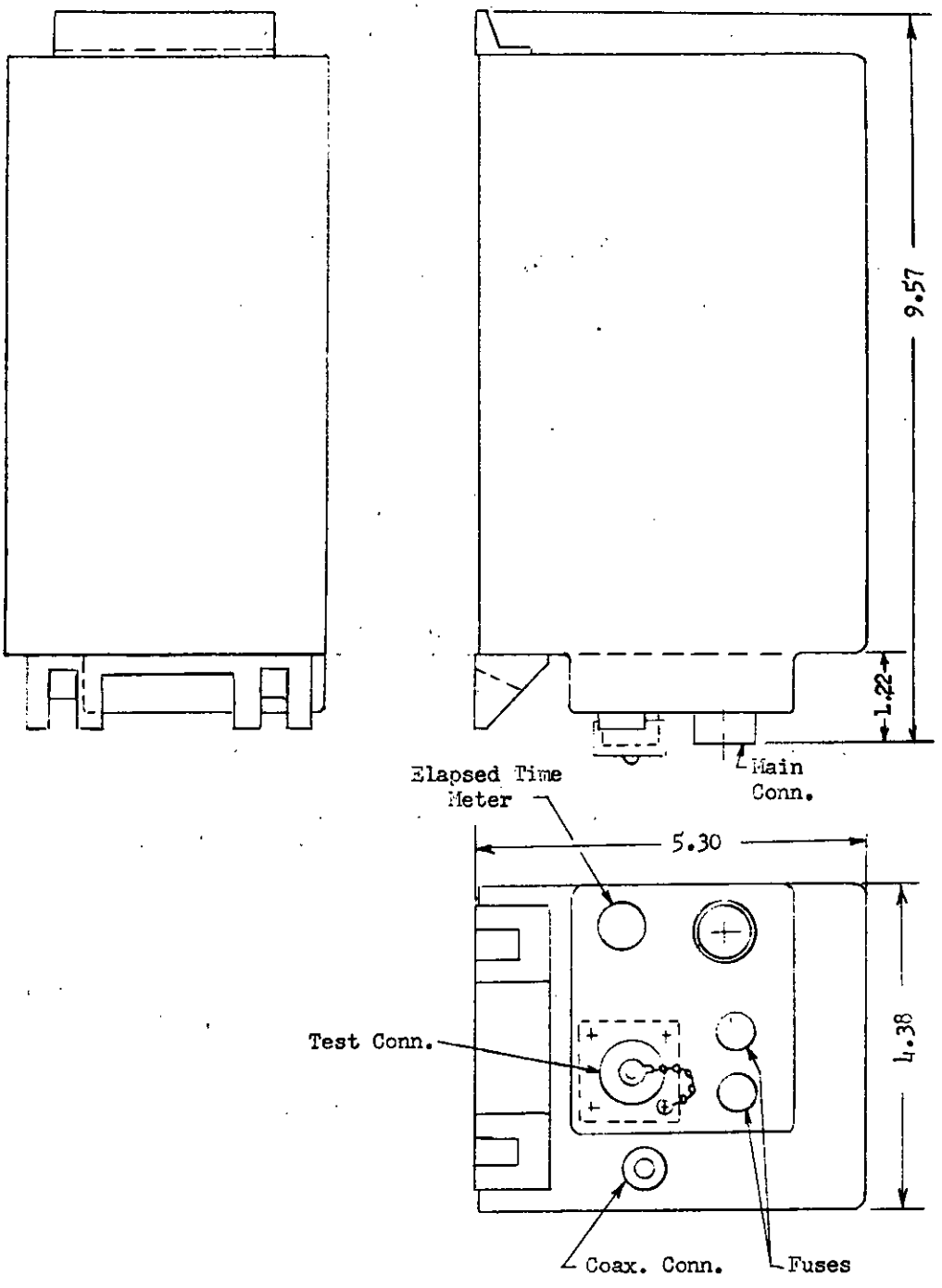


FIGURE 1. Receiver, Radio R-1286/ARR-69 and R-1286A/ARR-69A

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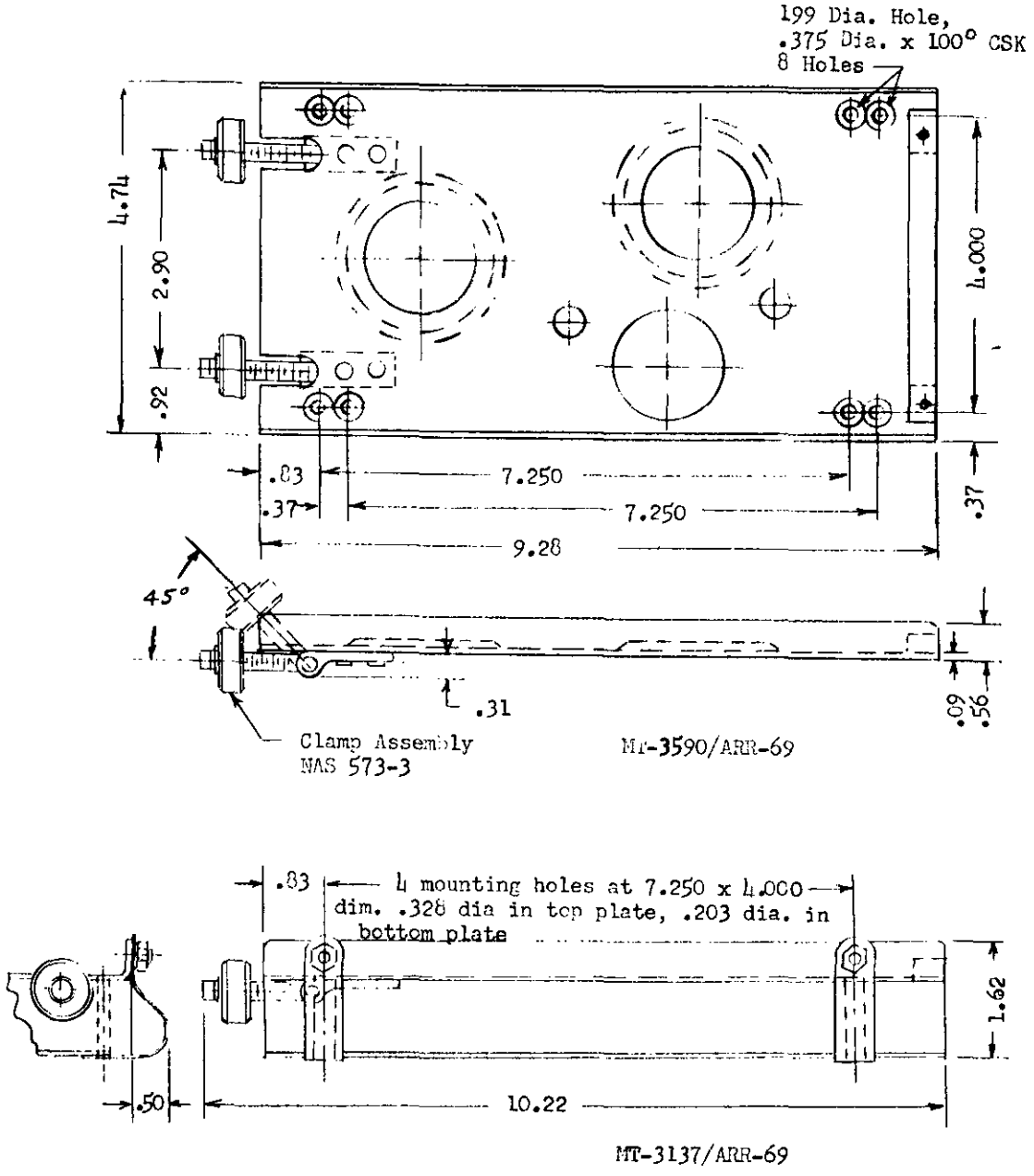


FIGURE 2. Mountings

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-23767B(AS) RECEIVING SET, RADIO AN/ARR-69 and AN/ARR-69A

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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